

**Example 3:** There is nothing that prevents a set from having seemingly unrelated elements. For instance,  $\{a, 2, \text{Fred}, \text{New Jersey}\}$  is the set containing the four elements  $a, 2, \text{Fred}$  and  $\text{New Jersey}$ .  $\square$

Sometimes the roster method is used to describe a set without listing all its members. Some members of the set are listed, and then ellipses ( $\dots$ ) are used when the general pattern of the elements is obvious.

**Example 4:** The set of positive integers less than 100 can be denoted by  $\{1, 2, 3, \dots, 99\}$ .  $\square$

### Set Builder Notation

- The general form of this notation is  $\{x \mid x \text{ has property } P\}$  and is read "the set of all  $x$  such that  $x$  has property  $P$ ."
- For instance, the set  $O$  of all odd positive integers less than 10 can be written as  $O = \{x \mid x \text{ is an odd positive integer less than } 10\}$ .
- or specifying the universe as the set of positive integers, as  $O = \{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 10\}$ .

We often use this type of notation to describe sets when it is impossible to list all the elements of the set.

For instance, the set  $\mathbb{Q}^+$  of all possible rational numbers can be written as  $\mathbb{Q}^+ = \{x \in \mathbb{R} \mid x = \frac{p}{q} \text{ for some positive integers } p \text{ and } q\}$ .

### Some important sets

1.  $N = \{0, 1, 2, 3, \dots\}$ , the set of all natural numbers
2.  $Z = \{\dots, -2, -1, 0, 1, 2, \dots\}$ , the set of all integers
3.  $Z^+ = \{1, 2, 3, \dots\}$ , the set of all positive integers
4.  $\mathbb{Q} = \{p/q \mid p \in Z, q \in Z, \text{ and } q \neq 0\}$ , the set of all rational numbers.
5.  $\mathbb{R}$ , the set of all real numbers.
6.  $\mathbb{R}^+$ , the set of all positive real numbers.